

Practical - 4

Practical Title: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

Theory :

1. How are DOM elements selected and modified?

DOM means Document Object Model which is very important concept in Web development also in JS . DOM manipulation means selecting an HTML element using JavaScript and then changing its text, HTML, style, attributes, classes, or values.

2. Are event listeners attached correctly?

Yes , In Java Script we use `addEventListener` so we use multiple event instead of single event handle by function so , event listeners are correctly attached using `addEventListener()`. It connects an HTML element with an event such as click or submit and executes a function when that event occurs.

3. Is `localStorage` used for remembering theme choice?

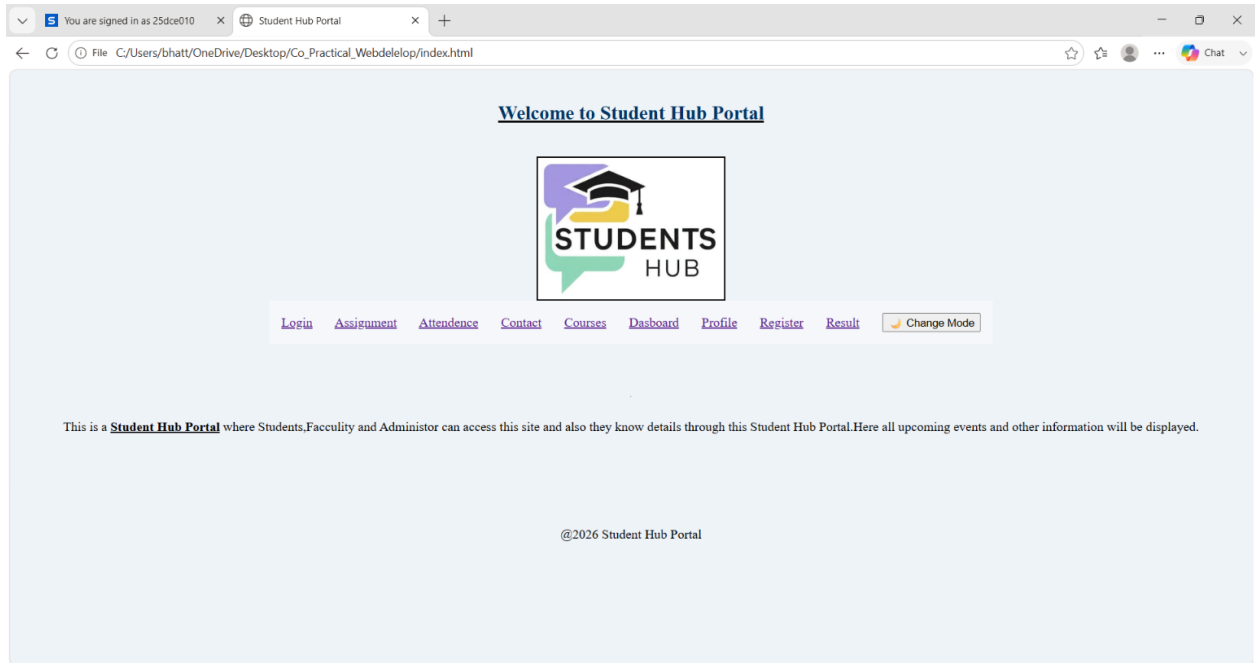
Yes. `localStorage` can be used to remember the user's theme choice. `setItem()` stores the selected theme, and `getItem()` retrieves it when the page is reopened, allowing the website to maintain the Dark/Light mode preference also other changes according to user requirement.

4. Does interactivity improve usability without breaking accessibility?

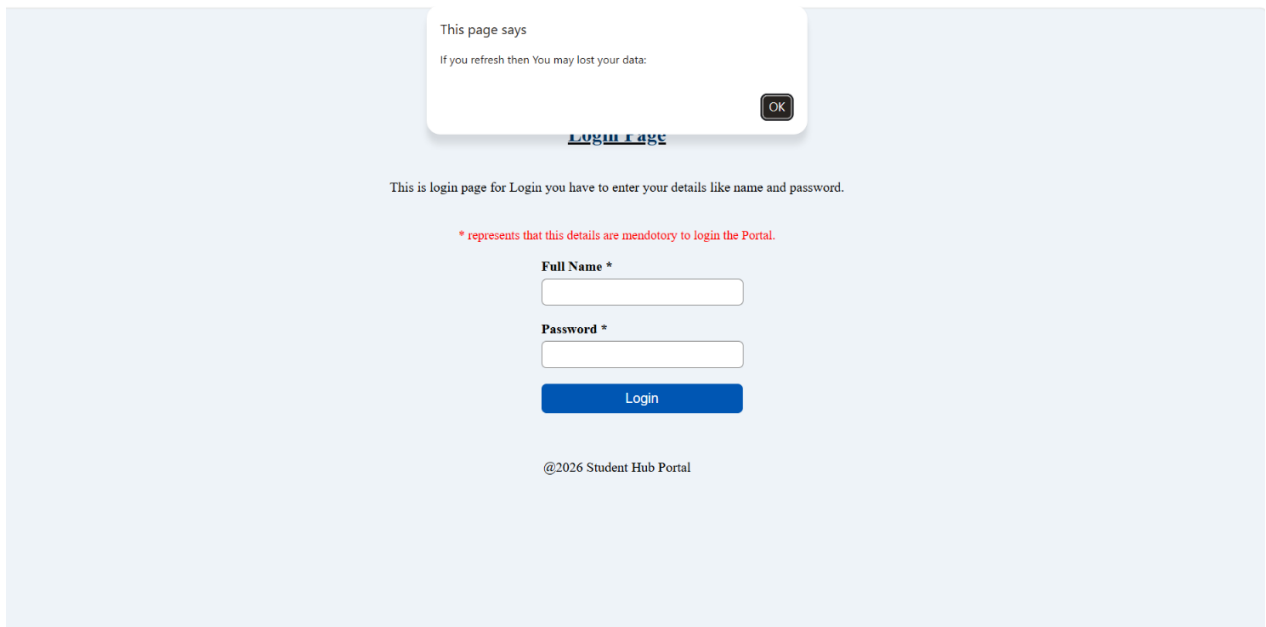
Yes. Interactivity improves usability when features such as buttons, notifications, menus, and forms make the website easier to use. Accessibility should be maintained by using proper HTML elements, keyboard support, readable text, clear messages, and sufficient contrast so all users can interact with the website.

Implementation/ Output/Result (Pages with CSS and JS)

(1)Home Page:



(2)Login Page:



(3)Try to feel with empty filled with blur event (both) JS works properly.

This page says
User doesn't fill fields:

OK

This is login page for Login you have to enter your details like name and password.

* represents that this details are mandatory to login the Portal.

Full Name *

Password *

Login

Please fill username

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(4)Login successful.

This page says
Your login is successful.

OK

This is login page for Login you have to enter your details like name and password.

* represents that this details are mandatory to login the Portal.

Full Name *

meet

Password *

....

Login

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Tool icons have been removed. The functionality is unchanged, and all tools are still available. If icons were important to your workflow, please share your feedback at <https://issues.chromium.org/issues/5255810>

Console Issues +

Filter

Default levels 1

Hello from console: javascript.js:3

Login successful: javascript.js:41

(5) User try to fill form with submit assignment and message form is display.

Assignment Page

This page is generally for assignment where Students can upload Assignments.
Here some are important instructions about Assignments.

Instructions :

- Submit before the deadline.
- Check the file before uploading.
- Keep a copy of your assignment.
- Contact faculty if you face any issue.

Assignment ID	Subject	Assignment Title	Faculty	Due Date	Status
A101	Web Development	Semantic HTML5 Pages	Prof. Patel	25 July 2026	Pending
A102	Java Programming	Methods and Classes	Prof. Shah	28 July 2026	Submitted
A103	Data Structures	Array Programs	Prof. Mehta	30 July 2026	Pending
A104	Computer Networks	Network Topologies	Prof. Joshi	02 August 2026	Submitted

Some Other Instructions are :

- Upload only PDF or DOC files.
- Maximum file size: 10 MB.
- Late submissions may receive reduced marks.
- Follow the prescribed format.

Select Option

Please select an assignment.

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(6) Successfully assignment submitted.

Assignment Page

This page is generally for assignment where Students can upload Assignments.
Here some are important instructions about Assignments.

Instructions :

- Submit before the deadline.
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Some Other Instructions are :

- Upload only PDF or DOC files.
- Maximum file size: 10 MB.
- Late submissions may receive reduced marks.
- Follow the prescribed format.

Assignment 1

Assignment 1 submitted successfully!

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(7) User try to fill leave application without selecting leave type.

Dashboard Page

In this page other details are provided like **Leave Application**, **Student Scholarship Details**, **Other Details** where Students can see multiple different things and select one form it.

Leave Application

Enter your details:

Full Name: *

Reason: *

Leave Type: *

--Select Leave Type--

Please select the leave type:

Submit

Student Scholarship Details:

Enter your details for scholarship registration

Tool icons have been removed. The functionality is unchanged, and all tools are still available. If icons were important to your workflow, please share your feedback at <https://issues.chromium.org/issues/52558>

Default levels: 3

Hello from console: `javascript.js:3`

User not select any `javascript.js:143` type of leave

(8) Leave type selected and successfully submitted .

Dashboard Page

In this page other details are provided like **Leave Application**, **Student Scholarship Details**, **Other Details** where Students can see multiple different things and select one form it.

Leave Application

Enter your details:

Full Name: *

Reason: *

Leave Type: *

Earned Leave

You have selected : earned

Student Scholarship Details:

Enter your details for scholarship registration

Full Name: *

Tool icons have been removed. The functionality is unchanged, and all tools are still available. If icons were important to your workflow, please share your feedback at <https://issues.chromium.org/issues/52558>

Default levels: 3

Hello from console: `javascript.js:3`

User selected leave `javascript.js:147` type is: earned

(9)User want to try to change some thing.

Profile Page

This page is generally for viewing and editing your profile information.

Student Profile

Student Full Name: Yash Rajubhai Aacharya

Student Id: 123

Student Department: Computer Engineering

Email ID: ****@gmail.com

Student Contact No: *****

Address: Gujarat, India

Semester: 3

Date of Birth: 15 August 2007

Gender: Male

Blood Group: O+

Meet Shaleshbhai Bhatti

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Tool icons have been removed. The functionality is unchanged, and all tools are still available. If icons were important to your workflow, please share your feedback at <https://issues.chromium.org/issues/52558101>

Elements +

Console Issues +

Filter

Default levels No Issues

Hello from console: javascript.js:3

(10)Successfully change and also mouseover event occurs.

Profile Page

This page is generally for viewing and editing your profile information.

Student Profile

Student Full Name: Meet Shaleshbhai Bhatti

Student Id: 123

Student Department: Computer Engineering

Email ID: ****@gmail.com

Student Contact No: *****

Address: Gujarat, India

Semester: 3

Date of Birth: 15 August 2007

Gender: Male

Blood Group: O+

Meet Shaleshbhai Bhatti

Name updated successfully.

Tool icons have been removed. The functionality is unchanged, and all tools are still available. If icons were important to your workflow, please share your feedback at <https://issues.chromium.org/issues/52558101>

Elements +

Console Issues +

Filter

Default levels No Issues

Hello from console: javascript.js:3

Name changed to: javascript.js:184

Meet Shaleshbhai Bhatti

GitHub Repository Link:

<https://github.com/MeetBhatti/WDF.git>

Learning Outcomes:

- Use **JavaScript DOM manipulation**
- Handle **user events** like clicks and form submissions
- Create interactive UI components
- Implement **dark/light theme switching**
- Use **localStorage** to remember user preferences
- Improve the website's **usability and interactivity**.

Faculty Remarks: (Keep this blank for faculty)